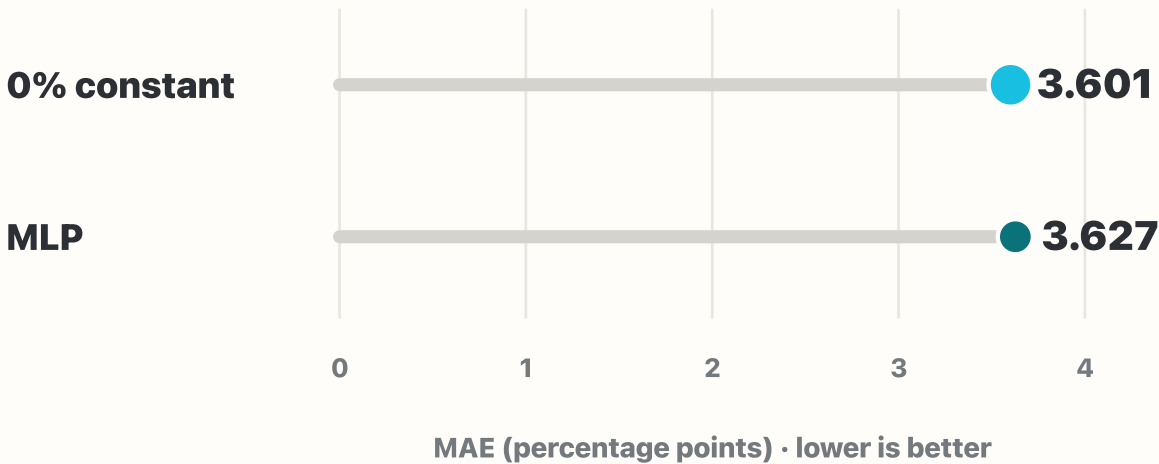
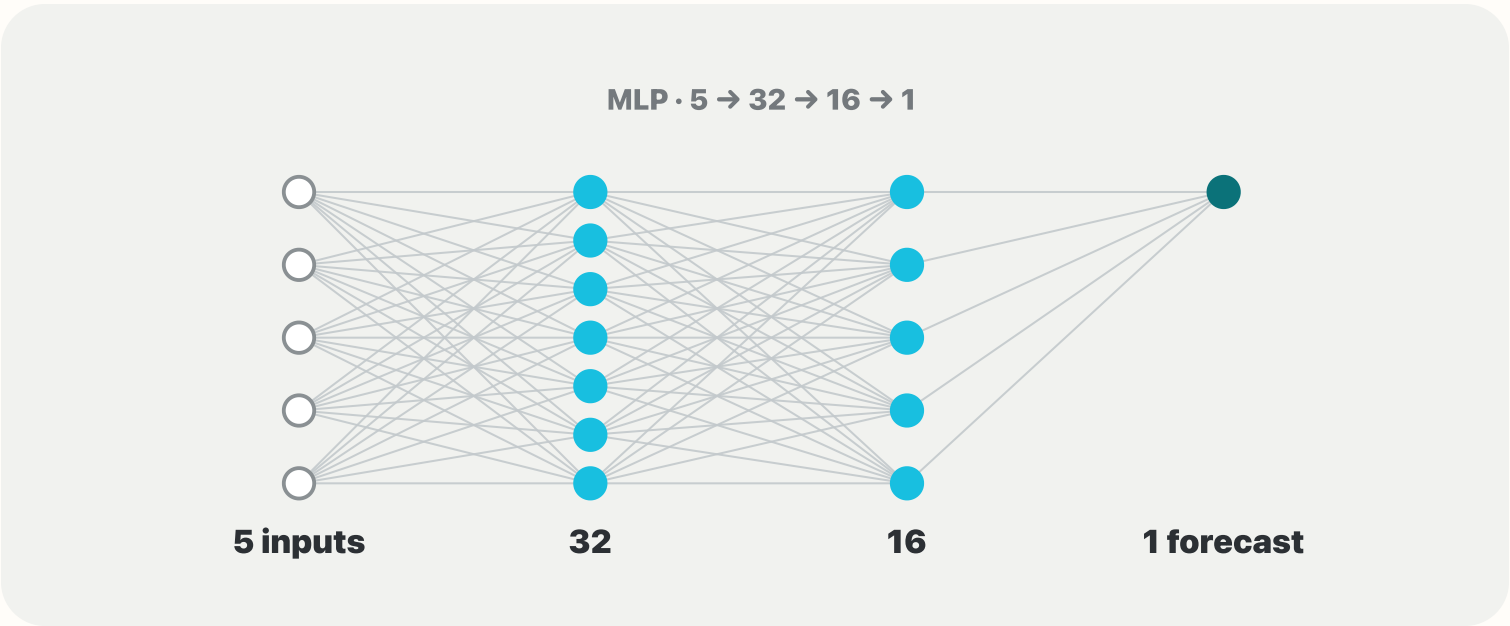


I SPENT 24 HOURS ON A NEURAL NETWORK. ALL I GOT WAS THIS LOUSY T-SHIRT.

My fancy model can learn very complex functions.
“Predict 0% growth” still beat it on average absolute error (MAE).

TARGET: each bank’s percentage change in domestic deposits next quarter.
\$100 million → \$96 million means **−4% actual growth**.
Predict 0%? That is a **4-percentage-point error**.



THE CONSTANT PREDICTED **0% GROWTH** FOR EVERY BANK.

DEPOSITS CAN LEAVE BEFORE LOANS PAY BACK.

EXAMPLE: \$100 IN ASSETS. \$10 CASH. \$90 LOANS.



CASH IN THE VAULT

\$10



MONEY ALREADY OUT AS LOANS

\$90

A customer withdraws \$15. Cash is \$5 short.

THE BANK'S TREASURY TEAM MUST FIND THE CASH.

WHO WATCHES THE BANK?



Bank managers manage cash.

Regulators oversee the bank.

Quarterly Call Reports show deposits, loans
and cash—the data I use here.

CASH

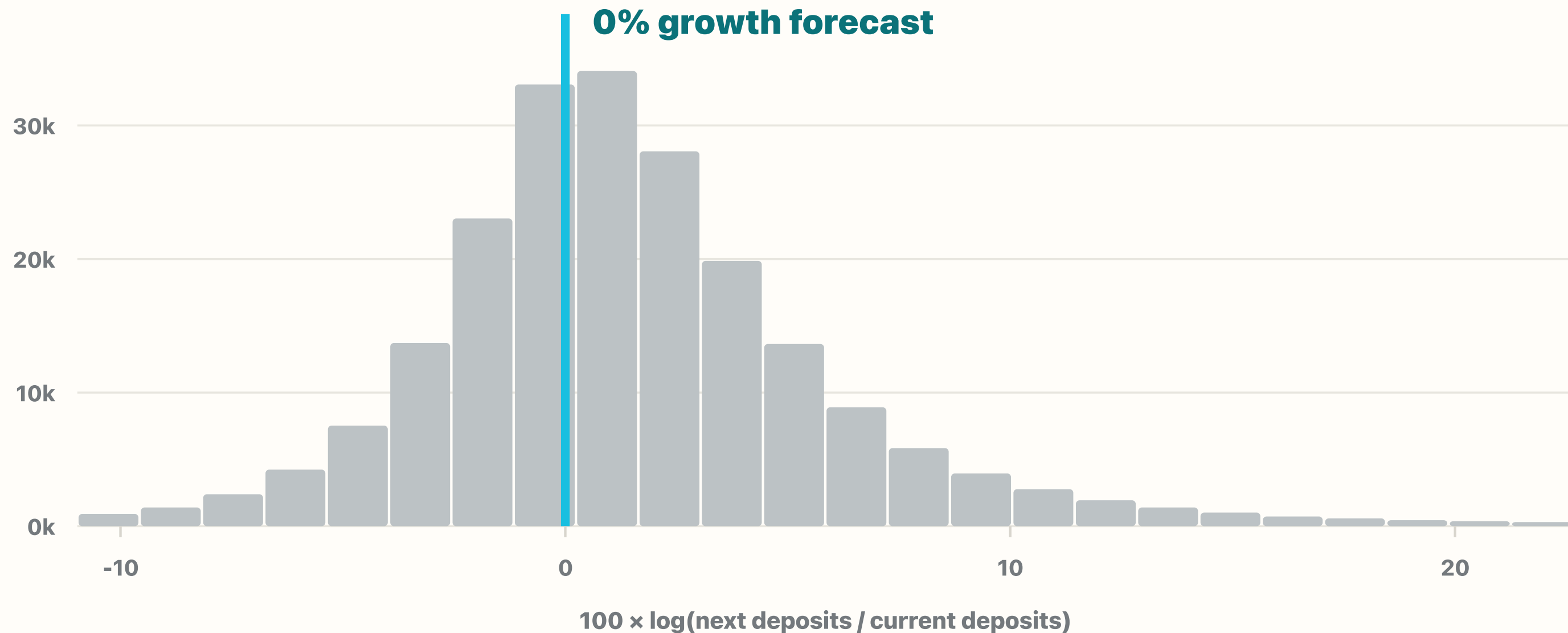
DEPOSITS

LOANS

CAPITAL

**CUSTOMERS DEPOSIT AND WITHDRAW MONEY.
I'M FORECASTING THE NET CHANGE NEXT QUARTER.**

SO WHY WAS 0% GROWTH SO HARD TO BEAT?



A LOT OF THE TRAINING TARGETS LIVE NEAR ZERO.

Central 98% shown · full tails stayed in the experiment · display scale is 100 × log growth

BIG ERRORS GET **SQUARED**. SO THEY CAN PULL THE MODELS AWAY FROM 0%.

An error 5× as large gets 25× the squared-error penalty.

FIXED BENCHMARK

0% BASELINE

$$\hat{y} = 0$$

Always predicts 0%.
It never adjusts to reduce an error.

ILLUSTRATIVE TRAINING ERRORS

2-unit error $\rightarrow 2^2$

10-unit error $\rightarrow 10^2$

4

VS

100

BIG ERRORS COST MUCH MORE.

LEARNED MODELS

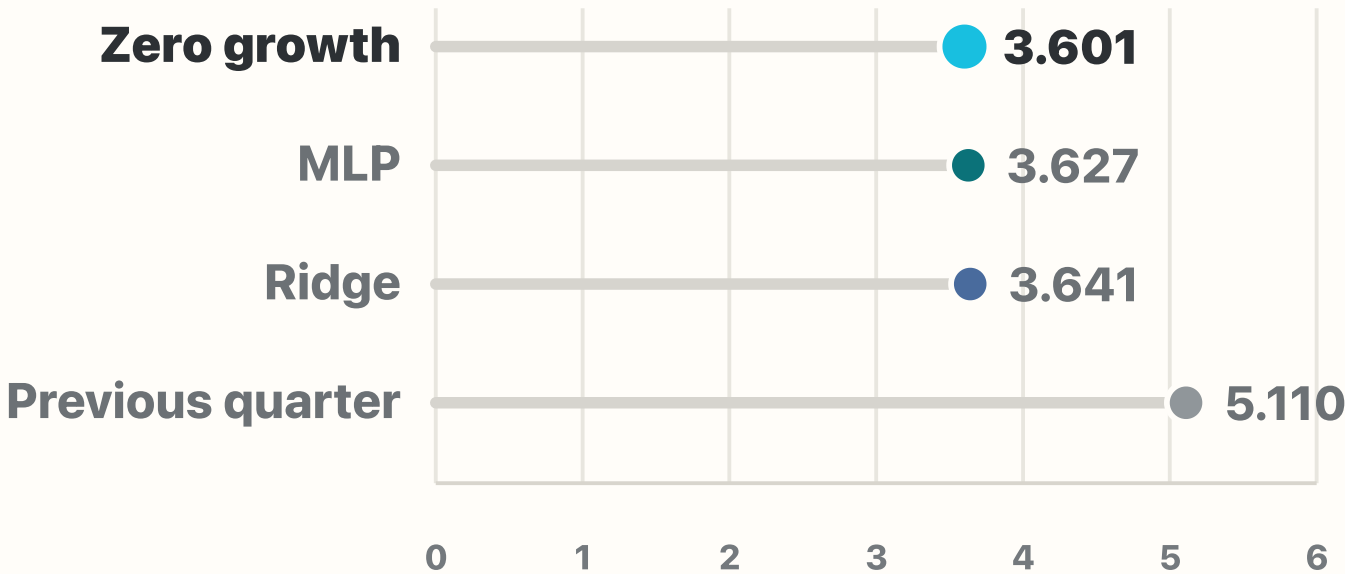
RIDGE + NEURAL NET

Can move away from 0% to reduce
those expensive large training errors.

**A LOWER SQUARED ERROR CAN COME WITH
A SLIGHTLY HIGHER AVERAGE ABSOLUTE ERROR.**

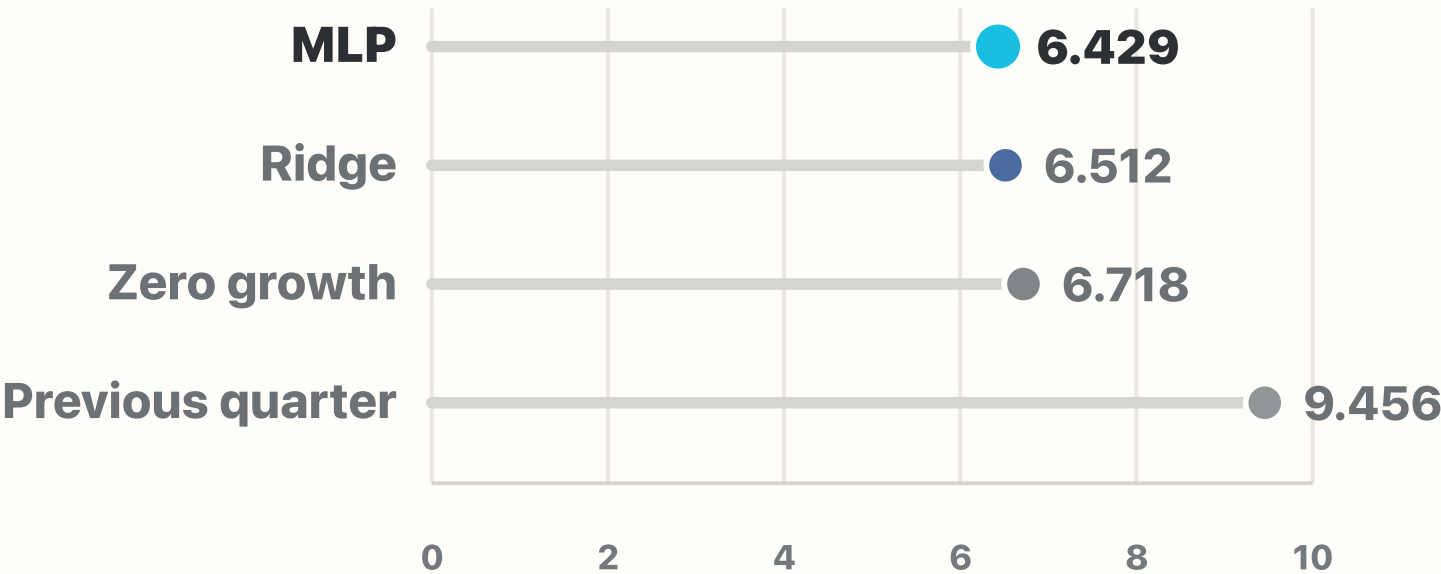
ZERO WON MAE. THE NEURAL NETWORK WON RMSE.

AVERAGE ABSOLUTE ERROR MAE



MAE (percentage points) · lower is better

LARGE ERRORS COUNT MORE RMSE



RMSE (percentage points) · lower is better

“WINNER” DEPENDS ON THE ERROR METRIC WE CHOOSE.
MAE averages absolute errors. RMSE squares errors first, so large errors count more.

0% TIES EVERY BANK.

THE LEARNED MODELS TELL ME WHO TO REVIEW FIRST.

In this March 2024 queue, review Young Americans Bank first: Ridge predicts the largest deposit decline, −4.02%. Check its reports to understand the forecast; this is not a current distress warning.

ORDERED REVIEW QUEUE · SIMPLE LINEAR MODEL (RIDGE) · FIRST 10 OF 454			
RANK	BANK	PREDICTED GROWTH	0% BASELINE
01	YOUNG AMERICANS BANK	-4.02%	0% · tied
02	PIONEER COMMERCIAL BANK	-3.32%	0% · tied
03	SUMITOMO MITSUI TR BK U SA	-2.83%	0% · tied
04	OLD GLORY BANK	-2.75%	0% · tied
05	TURTLE MOUNTAIN STATE BANK	-2.11%	0% · tied
06	COLUMBUS STATE BANK	-2.05%	0% · tied
07	FDS BANK	-2.03%	0% · tied
08	BOELUS STATE BANK	-1.96%	0% · tied
09	LEMONT NATIONAL BANK	-1.80%	0% · tied
10	CITIZENS NB OF CROSBYTON	-1.75%	0% · tied

HOW MANY OF THE ACTUAL 454 LOWEST-GROWTH BANKS MADE THE LIST?

Random selection

45.4 expected

Linear model (Ridge)

112 found

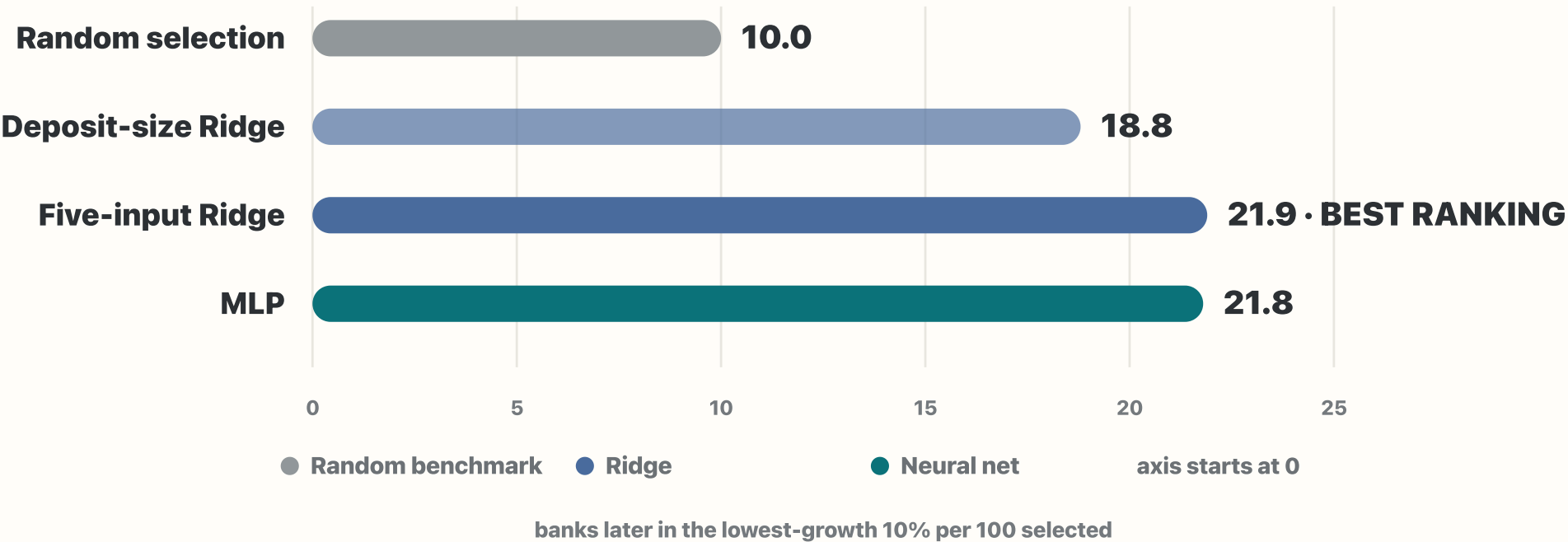
Neural network

109 found

4,536 eligible banks · March → June 2024. “Lowest-growth” = later in that quarter’s bottom 10% of deposit growth. 112 matches across the full linear-model queue; 109 for the neural net. Review priority is not a distress diagnosis.

I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.

I tested deposit size alone to see whether four extra inputs actually improved the review list.



MAE WINNER: PREDICT 0% → 3.601. Deposit-size Ridge: 3.654 · Five-input Ridge: 3.641 · Neural net: 3.627.
All three lost to zero on MAE. Zero ties every bank, so it cannot order a review list.

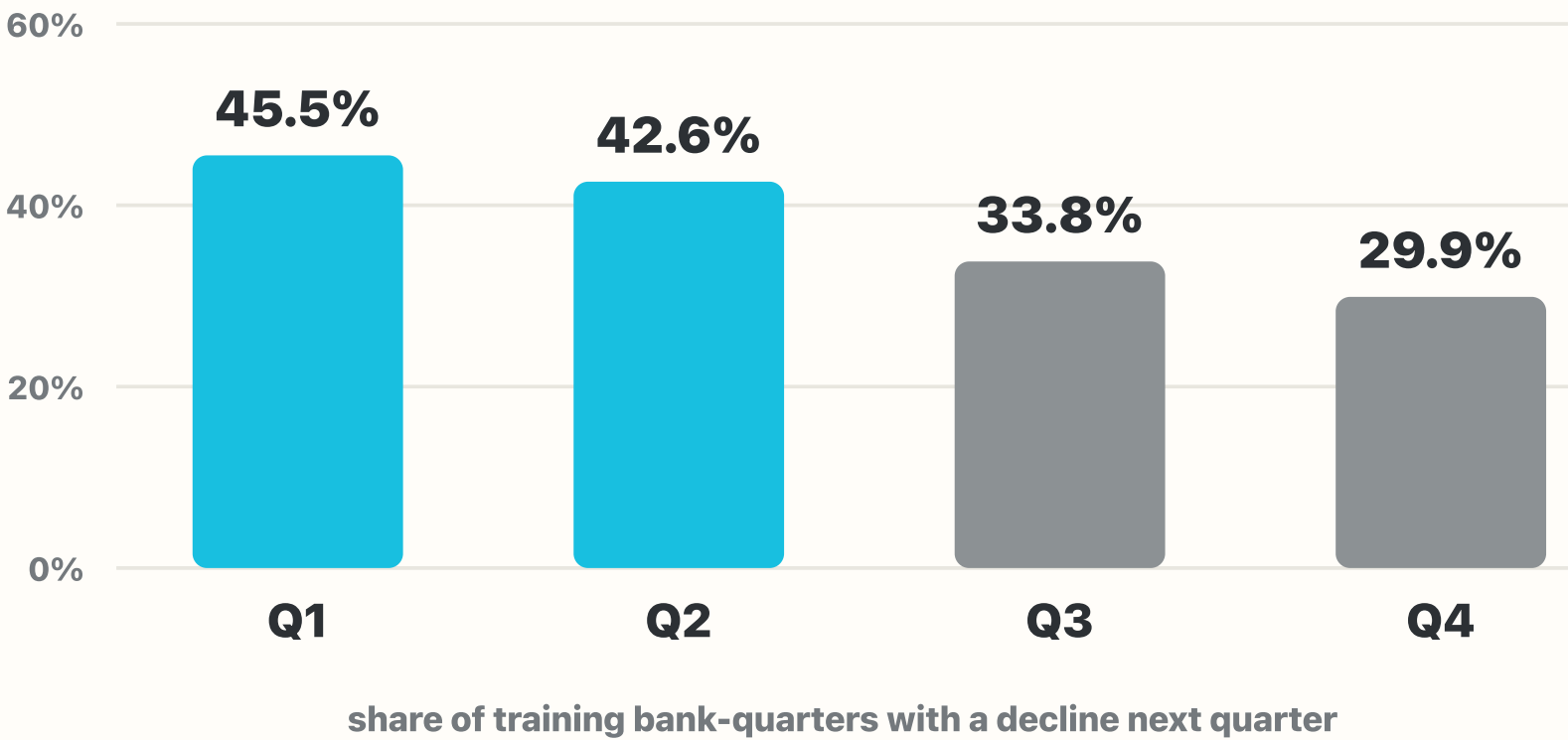
DEPOSIT SIZE ALONE: 18.8 FOUND PER 100 REVIEWED.
ALL FIVE INPUTS: 21.9. A GAP OF 3.1.

DEPOSIT DECLINES WERE MORE COMMON AFTER Q1 THAN Q4.

Q1: 45.5% of training bank-quarters were followed by a deposit decline. **Q4: 29.9%.**

That is how often declines happened, not “banks lost 45.5% of their deposits.”

TRAINING DATA, 2013–2022 · BANK-QUARTERS POOLED BY STARTING QUARTER



THE STARTING QUARTER MAY HELP SET A GROWTH FORECAST.

Each bar pools all eligible bank-quarter rows for that starting quarter in the training set (through Sep 2022). Declines ÷ rows; not an equal-weight average of yearly rates.

0% IS STILL THE BENCHMARK TO BEAT. DOES THE CALENDAR RULE EARN ITS PLACE?

June 30 → September 30, 2026 · one forecast per bank · fixed September 22

CALENDAR RULE · Q2 STARTING QUARTER

+0.550%

Same forecast for every bank. If actual growth is 0%, this misses by 0.550 pp.

NO-CHANGE BASELINE

0%

Predict no change. If actual growth is 0%, the miss is zero.

1. WAIT FOR SEPTEMBER REPORTS.

Actual growth =
 $100 \times (\text{September deposits} \div \text{June deposits} - 1)$.

2. MEASURE EVERY MISS.

Absolute miss = $|\text{actual growth} - \text{forecast}|$.
Example: +1.00% actual → 0.450 pp calendar miss; 1.000 pp zero miss.

3. COMPARE THE AVERAGES.

Lower MAE wins this quarter.
If the calendar rule loses, report the loss. Inspect misses before changing the rule.

**IF DEPOSITS DON'T CHANGE, ZERO IS EXACT.
THE CALENDAR RULE HAS TO EARN THE IMPROVEMENT.**

As of Sep 22, 2026: June data are listed in FDIC BankFind; the Sep 30 quarter has not ended. Saved rule = Q2 training median (+0.550399842%); developed after seeing 2024. Score matched banks with valid positive June and September domestic deposits; disclose missing reports and merger cases. Example is illustrative. One quarter is not lasting proof.